Synchronization Intro

- This week
  - Background
  - The Critical-Section Problem
  - User Library Support for Synchronization
    - Mutex Locks
    - Semaphores
    - Monitors

Synchronization Outline

- This week
  - Background
  - The Critical-Section Problem
  - User Library Support for Synchronization

Next week: Low-level Algorithmic Solutions

- C11 Atomic operations library
- Implementation of locks
  - kernel space
  - user level implementation
- Cache coherence
- Lock “Free” Multithreading
  - memory barriers
- Lock free data structures
  - RCU
- transactions

Next next week

- Review and summary of synchronization

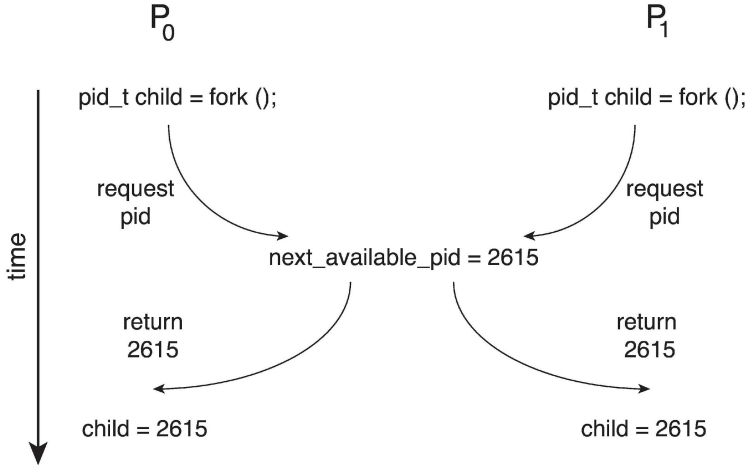
```
void *producer(void *data){
   while (1)    {
        /* produce an item in next produced
*/
        while (count == BUFFER_SIZE)
            ; /* do nothing */

        buffer[in] = produced;
        in = (in + 1) % BUFFER_SIZE;
        count++;
   }
}
```

```
void *consumer(void *data){
   while (1){

        while (count == 0)
            ; /* do nothing */

        consumed = buffer[out];
        out = (out + 1) % BUFFER_SIZE;
        count--;

        /* consume the item in next
consumed */
   }
}
```

```
// UNSYNCHRONIZED - THIS CODE HAS A RACE CONDITION
```

Race Condition Example in Kernel

- Processes P_0 and P_1 are creating child processes using the `fork()` system call
- Race condition on kernel variable `next_available_pid`which represents the next available process identifier (pid)



- Unless there is a mechanism to prevent P_0 and P_1 from accessing the variable `next_available_pid` the same pid could be assigned to two different processes!

Explicit Example in User program

```c
#include <stdio.h>
#include <pthread.h>

int sum = 0; //shared

void *countgold(void *param) {
 int i; //local to each thread
 for (i = 0; i < 10000000; i++) {
   sum += 1;
 }
 return NULL;
}
```

```c
int main() {
 pthread_t tid1, tid2;
 pthread_create(&tid1, NULL, countgold, NULL);
 pthread_create(&tid2, NULL, countgold, NULL);

 //Wait for both threads to finish:
 pthread_join(tid1, NULL);
 pthread_join(tid2, NULL);

 printf("ARRRRG sum is %d\n", sum);
 return 0;
}
```

```
$ gcc main.c -pthread
$ ./a.out
??????????
```

When multithreading gets “interesting”!

```c
#include <stdio.h>
#include <pthread.h>

int sum = 0; //shared

void *countgold(void *param) {
 int i; //local to each thread
 for (i = 0; i < 10000000; i++) {
   sum += 1;
 }
 return NULL;
}
```

```c
int main() {
 pthread_t tid1, tid2;
 pthread_create(&tid1, NULL, countgold, NULL);
 pthread_create(&tid2, NULL, countgold, NULL);

 //Wait for bo
 pthread_join
 pthread_join

 printf("ARRRR
 return 0;
}
```

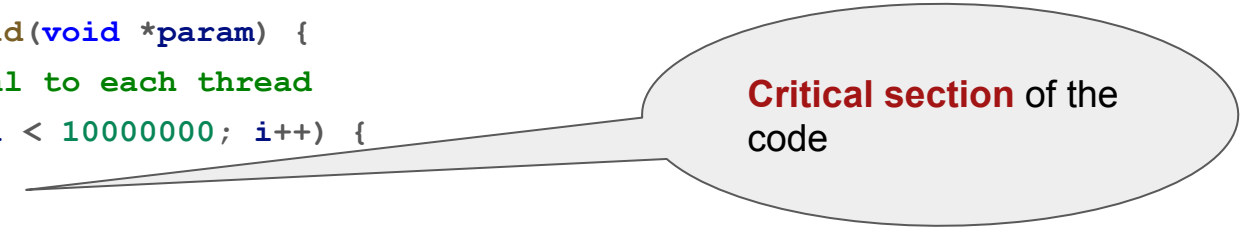
```
$ gcc main.c -pthread
$ ./a.out
ARRRRG sum is 10214142
$ ./a.out
ARRRRG sum is 11798888
$ ./a.out
ARRRRG sum is 10576703
$ ./a.out
ARRRRG sum is 10159994
```

There is a race condition if the end result may change based on the execution order.

Mutual exclusion

```
From previous example
#include <stdio.h>
#include <pthread.h>

int sum = 0; //shared

void *countgold(void *param) {
 int i; //local to each thread
 for (i = 0; i < 10000000; i++) {
  sum += 1;
 }
 return NULL;
}
```

We do not want two or more threads to update `sum` at the same time.



Mutual exclusion

Mutual exclusion is a property that says

“no two or more processes (threads) can be in their critical section at the same time at any given point of time”.

Critical Section

- General structure of process P_i

```
while (true) {
        entry section
            critical section
        exit section
            remainder section
}
```

Requirements for solution to critical-section problem

1. **Mutual Exclusion**
   - If process P_i is executing in its critical section, then no other processes can be executing in their critical sections
2. **Progress**
   - If no process is executing in its critical section and there exist some processes that wish to enter their critical section, then the selection of the process that will enter the critical section next cannot be postponed indefinitely
3. **Bounded Waiting**
   - A bound must exist on the number of times that other processes are allowed to enter their critical sections after a process has made a request to enter its critical section and before that request is granted

Hardware solutions:
Interrupt-based solution

- Entry section: disable interrupts
- Exit section: enable interrupts

```
while (true) {
        entry section
            critical section
        exit section
            remainder section
}
```

- Will this solve the problem?
  - What if the critical section-code runs for an hour?
  - Can some processes starve
    - never enter their critical section.
  - What if there are two CPUs?

Software-user library solutions

Thread packages typically provide

mutex variables

- acquire-release
- lock-unlock

semaphores (through system calls)

Pthread (POSIX Threads)Libraries for User Apps

Thread packages typically provide mutexes:

```
void mutex_init (mutex_t *m, ...);
void mutex_lock (mutex_t *m);
int mutex_trylock (mutex_t *m);
void mutex_unlock (mutex_t *m);
```

- Only one thread acquires m at a time, others wait

Mutual exclusion

Mutual exclusion is a property that says

“no two or more processes (threads) can be in their critical section at the same time at any given point of time”.

Thread synchronization

Mutex ("mutual exclusion")

- Protect accesses to shared variables at the same time

Condition Variables

- Inform another thread about the change in the state of the shared variable
  - synchronize based upon the actual value of the variable

Mutex variables

To achieve mutual exclusion we use mutex variable

It ensures that only one thread at a time can access a global variable

In POSIX: **pthread_mutex_t,**

similar lock concepts in different languages

```
/* global variable should be used for
static mutexes*/
pthread_mutex_t m;
m = PTHREAD_MUTEX_INITIALIZER;

/* start of Critical Section */
pthread_mutex_lock(&m);

// Critical section

/* end of Critical Section */
pthread_mutex_unlock(&m);
```

Mutexes

```
//m1 is a mutex variable
mutex_lock(m1);//acquire lock
critical_section
mutex_unlock(m1); //release lock
```

- You can consider lock as **“a mic among participants that controls who has the right to speak”**,
  - That is whoever has the mic (m1) has the right to speak
    - in our case do ops on the memory shared among the participants.

Mutex variables

- A mutex can either be allocated as a static (global) variable
- or be created dynamically at run time

```
pthread_mutex_t m = PTHREAD_MUTEX_INITIALIZER;
pthread_mutex_lock(&m);
// Critical section
pthread_mutex_unlock(&m);
```

```
pthread_mutex_t *lock = malloc(sizeof(pthread_mutex_t));
pthread_mutex_init(lock, NULL);
//critical section
pthread_mutex_destroy(lock);
free(lock);
```

Example Sum

```
/* Create a mutex this ready to be locked!*/
pthread_mutex_t m = PTHREAD_MUTEX_INITIALIZER;
int sum = 0;
void *countgold(void *param) {

    pthread_mutex_lock(&m);
    /* Other threads that call lock will have to wait until we call unlock */
    for (int i = 0; i < 10000000; i++) {
        sum += 1; /*->critical section*/
    }
    pthread_mutex_unlock(&m);

    return NULL;
}
```

- Same thread that locks the mutex must unlock it
- Critical section is 'sum += 1'

Which one runs faster?

```
void *countgold(void *param){
 int i;
 for (i = 0; i < 10000000; i++){
   pthread_mutex_lock(&m);
   sum += 1;
   pthread_mutex_unlock(&m);
 }
 return NULL;
}
```

```
void *countgold(void *param){
 int i, local = 0;
 for (i = 0; i < 10000000; i++) {
   local += 1;
 }

 pthread_mutex_lock(&m);
 sum += local;
 pthread_mutex_unlock(&m);

 return NULL;
}
```

This runs slower because we lock and unlock the mutex a million times, which is expensive - at least compared with incrementing a variable.

Example: sum of elements in array

Write a concurrent program to find the sum of elements in array

- Define a global sum
- Each thread sums elements between start and end
  - Start & end are computed from thread index
- Adds its resulting sum to the global sum

Some gotchas!

```
int a;
pthread_mutex_t m1 = PTHREAD_MUTEX_INITIALIZER,
                m2 = PTHREAD_MUTEX_INITIALIZER;
// later
// Thread 1
pthread_mutex_lock(&m1);
a++;
pthread_mutex_unlock(&m1);

// Thread 2
pthread_mutex_lock(&m2);
a++;
pthread_mutex_unlock(&m2);
```

This code creates a mutex that does effectively nothing!

Condition variables: Overview

mutexes

- implement synchronization by controlling thread access to data,
- prevent multiple threads accessing the shared variable at the same time.
  - Threads must be participating

condition variables

- allow threads to synchronize based upon the actual value of data.
  - one thread informs other threads about changes in the state of a shared variable (or other shared resource)
  - and allows other threads to wait (block) for such notification.
- allow a set of threads to sleep until woken up.
- A condition variable is always used in conjunction with a mutex lock

Monitor

a monitor is

- a synchronization construct
- “Pattern”

A monitor consists of

- a mutex (lock)
  - It uses the **lock** to protect a critical section.
- and at least **one condition variable**.
  - allows them to wait for the state to change

It provides a mechanism for threads to temporarily give up exclusive access in order to wait for some condition to be met

A basic monitor construct

```
lock(m);                    // Acquire this monitor's lock.
/*... Critical section ...*/

while (!p) {            // While the condition that we are waiting  for is not true
   wait(cv, m);         // Wait on this monitor's lock and condition variable.
}

/*... Critical section ...*/
signal(cv2);                    // Or: broadcast(cv2);
                                // cv2 might be the same as cv or different.
unlock(m);                      // Release this monitor's lock.
```

Monitor for waiting two different conditions

```
//1st thread
mutex_lock(m1);

/*critical section entry*/
while(need_to_wait_1 ){
   cond_wait(c1, m1);
}

/*critical section exit*/
cond_signal(c2) //or broadcast
mutex_unlock(m1);
```

```
// 2nd thread
mutex_lock(m1);

/*critical section entry*/
while(need_to_wait_2){
   cond_wait(c2, m1);
};

/*critical section exit*/
cond_signal(c1) //or broadcast
mutex_unlock(m1);
```

- **m1:** mutex lock to control the access to the critical section.
- **c1: condition variable** is for one condition
- c2: **condition variable** is for another condition

Creating condition variables

Condition variables must be declared with type **pthread_cond_t**, and must be initialized before they can be used. There are two ways to initialize a condition variable:

Statically,.

```
pthread_cond_t cond = PTHREAD_COND_INITIALIZER;
```

Dynamically,

```
int pthread_cond_init(pthread_cond_t *cond, pthread_condattr_t * attr);
```

- The ID of the created condition variable is returned to the calling thread through the condition parameter.
  - This method permits setting condition variable object attributes, attr.

```
int pthread_cond_destroy(pthread_cond_t *cond);
```

Signaling and waiting on condition variables

```
#include <pthread.h>
int pthread_cond_wait(pthread_cond_t *cond, pthread_mutex_t *mutex);
int pthread_cond_signal(pthread_cond_t *cond);/*only one waiting thread is woken*/
int pthread_cond_broadcast(pthread_cond_t *cond); /*all waiting threads are woken*/
```

All return 0 on success, or a positive error number on error

Condition variables are always used with a mutex lock.

Before calling wait, the mutex lock must be locked and wait must be wrapped with a loop.

```
pthread_mutex_lock();
    while(condition_is_false)
        pthread_cond_wait();
pthread_mutex_unlock();
```

A **while loop** is used because the condition could change after `pthread_cond_wait()`.

- Wait wakes the thread but the state of the buffer might change until the thread gets to really run.

producer-consumer example

```
pthread_mutex_t mtx =
    PTHREAD_MUTEX_INITIALIZER;
int avail = 0;
void* producer(void * p){
   pthread_mutex_lock(&mtx);
   avail++;
   pthread_mutex_unlock(&mtx);
 }
```

```
int main(){
   for (;;) {
       pthread_mutex_lock(&mtx);
       while (avail > 0) {
           avail--;
       }
       pthread_mutex_unlock(&mtx);
   }
}
```

The above code works, but it wastes CPU time, because the main thread continually loops, checking the state of the variable avail.

producer-consumer example with cond variable

```
pthread_mutex_t mtx =
   PTHREAD_MUTEX_INITIALIZER;
int avail = 0;
void *producer(void *p) {
 pthread_mutex_lock(&mtx);
 avail++;
 pthread_cond_signal(&cv);
 pthread_mutex_unlock(&mtx);
}
```

```
void *consumer(void *p) {
 for (;;) {
   pthread_mutex_lock(&mtx);
   while (avail == 0) {
     pthread_cond_wait(&cv, &mtx);
   }
   /*consume*/
   while (avail > 0) {
     avail--;
   }
   pthread_mutex_unlock(&mtx);
 }
}
```

A condition variable allows a thread to sleep (wait) until another thread notifies (signals) it that it must do something

Details of wait

`pthread_cond_wait(&cv, &mtx)`

- unlocks the mutex
- and begins waiting for the condition variable to be signalled.

```
pthread_mutex_lock(&mtx);
while (avail == 0) {
    pthread_cond_wait(&cv, &mtx);
}
```

- before invoking wait(), you must have ownership of **mtx**
- pthread_cond_wait() returns with the mutex **locked! You must unlock at the end!**

```
pthread_mutex_unlock(&mtx);
```

If there is no-space, to prevent overflow:

★ **You can also add condition to producer.**

The main design is the following:

```
//1st thread
mutex_lock(m1);
/*critical section entry*/
while(need_to_wait_1 ){
   cond_wait(c1, m1);
}

/*critical section exit*/
cond_signal(c2) //or broadcast
mutex_unlock(m1);
```

```
// 2nd thread
mutex_lock(m1);
/*critical section entry*/
while(need_to_wait_2){
   cond_wait(c2, m1);
};

/*critical section exit*/
cond_signal(c1) //or broadcast
mutex_unlock(m1);
```

Producer-consumer with overflow control

```
pthread_mutex_t m = PTHREAD_MUTEX_INITIALIZER;
pthread_cond_t notempty = PTHREAD_COND_INITIALIZER;
pthread_cond_t notfull = PTHREAD_COND_INITIALIZER;
int count = 0;
int buffer[BUFFER_SIZE];
```

```
void *producer(void *ptr) {

   pthread_mutex_lock(&m);
   while (count >= BUFFER_SIZE)
     pthread_cond_wait(&notfull, &m);

   /*produce 1 item*/
   printf("producing item:%d", count);
   buffer[count] = rand();
   count = count + 1;

   pthread_cond_signal(&notempty);
   pthread_mutex_unlock(&m);
}
```

```
void *consumer(void *ptr) {
 // for(;;){
 pthread_mutex_lock(&m);
 while (count == 0)
   pthread_cond_wait(&notempty, &m);

 /*consume 1 item*/
 printf("consumed item %d - %d\n",
        count, buffer[count]);
 count = count - 1;

 pthread_cond_signal(&notfull);
 pthread_mutex_unlock(&m);
}
```

What if we have more than one shared resources?

- use many mutexes?

What about processes?

Semaphores: Overview

```
sem_t s;
sem_init(&s, 0, 10); //WE HAVE 10 available resources (say 10-chairs)

//a thread that wants to occupy a chair
sem_wait(&s); //down the value of s by 1
 ...

//After done with the chair
sem_post(&s); //up the value of s by 1
```

- If the return value of `sem_wait` is negative the thread waits as done in the mutex locks.
  - This happens when the value of s before `sem_wait` is 0.
- `sem_wait` and `sem_post` can be called from different processes/threads.

POSIX Semaphores

```
int sem_init(sem_t *sem,
             int pshared,
             unsigned int value);
```

initializes the unnamed semaphore at the address pointed to by **`sem`**.

`value`

- specifies t**he initial value** for **`sem`**

`pshared`: indicates whether between process or threads

- 0 between threads
  - Should be visible to all threads,
    - Global variable
- Non-zero between processes

allow processes and threads to synchronize access to shared resources

A semaphore is a type of lockable object.

```
sem_t s;

sem_init(&s, 0, 10); /*returns -1 on error*/
```

Decrementing the semaphore value

```
sem_t s;

sem_init(&s, 0, 10); /*returns -1 on error*/
```

***decrementing* the value of semaphore:** You take ownership of a semaphore with a *wait* operation(abstractly called *P from Dijkstra's paper*),

```
sem_wait(&s);
```

- **Could do this 10 times without blocking**

POSIX Semaphores

***incrementing* the value of semaphore:**
You release ownership with a *signal* operation, a *post* operation, or abstractly called *V.*

sem_post(&s);

- **Announce that we've finished**
- **And one more resource item is available;**
- **increment the value of semaphore**

To destroy semaphore

- **release resources of the semaphore**

sem_destroy(&s);

.

Binary semaphore vs mutex

A binary semaphore is a semaphore with a maximum count of 1.

sem_t s;

sem_init(&s, 0, 1);

- A mutex is a semaphore that always **waits** before it **posts**.
- A binary semaphore can be used as a mutex by requiring that a thread only signals the semaphore (to unlock the mutex) if it was the thread that last successfully waited on it (when it locked the mutex).

- mutex ownership is tied to a thread, and only the thread that acquired the lock on a mutex can release it

- the semaphore doesn't care if the "wrong" thread signals the semaphore.

- sem_wait and sem_post can be called from different threads!

Semaphore vs mutex

```
1.  // Thread 1
2.  sem_wait(&s);
3.  // Critical Section
4.  sem_post(&s);
```

```
5.  // Thread 2
6.  // Some threads want to see the world burn
7.  sem_post(&s);
```

```
8.  // Thread 3
9.  sem_wait(&s);
10. // Not thread-safe!
11. sem_post(&s);
```

```
1.  // Thread 1
2.  mutex_lock(&s);
3.  // Critical Section
4.  mutex_unlock(&s);
```

```
5.  // Thread 2
6.  // Failed!
7.  mutex_unlock(&s);
```

```
8.  // Thread 3
9.  mutex_lock(&s);
10. // Now it's thread-safe
11. mutex_unlock(&s);
```

Making Data Structures Thread-Safe

```
int count = 0;
double values[STACK_SIZE];
void push(double v) {
   values[count++] = v;
}
double pop() {
   return values[--count];
}
int is_empty() {
   return count == 0;
}
```

it is multithread-unsafe:

- if two threads call push or pop at the same time then the results or the stack can be inconsistent.
  - both threads may read the original same count value.

Version-2 of stack

```
#define STACK_SIZE 20
int count;
double values[STACK_SIZE];
pthread_mutex_t m1 =
PTHREAD_MUTEX_INITIALIZER;
pthread_mutex_t m2 =
PTHREAD_MUTEX_INITIALIZER;
void push(double v) {
   pthread_mutex_lock(&m1);
   values[count++] = v;
   pthread_mutex_unlock(&m1);
}
```

```
double pop() {
   pthread_mutex_lock(&m2);
   double v = values[--count];
   pthread_mutex_unlock(&m2);
   return v;
}
int is_empty() {
   pthread_mutex_lock(&m1);
   return count == 0;
   pthread_mutex_unlock(&m1);
}
```

This version contains 2 errors!
What are the errors?

Version 3

```
// Another attempt to a thread-safe stack
#define STACK_SIZE 20
int count;
double values[STACK_SIZE];
pthread_mutex_t m1 =
PTHREAD_MUTEX_INITIALIZER;
void push(double v) {/*may be overflow*/
   pthread_mutex_lock(&m1);
   values[count++] = v;
   pthread_mutex_unlock(&m1);
}
```

```
double pop() {/*may be underflow*/
   pthread_mutex_lock(&m1);
   double v = values[--count];
   pthread_mutex_unlock(&m1);
   return v;
}
int is_empty() {
   pthread_mutex_lock(&m1);
   int r = count == 0;
   pthread_mutex_unlock(&m1);
   return r;/*may be out of date*/
}
```

- is_empty is thread-safe but its result may already be out-of-date. This is usually why in thread-safe data structures, functions that return sizes are removed or deprecated.
- There is no protection against underflow (popping on an empty stack) or overflow (pushing onto an already-full stack)

A general stack

A more general-purpose version might include the mutex as part of the memory structure and use pthread_mutex_init to initialize the mutex.

```
typedef struct stack {
    int count;
    pthread_mutex_t m;
    double *values;
} stack_t;

stack_t* stack_create(int capacity) {
    stack_t *result = malloc(sizeof(stack_t));
    result->count = 0;
    result->values = malloc(sizeof(double) * capacity);
    pthread_mutex_init(&result->m, NULL);
    return result;
}

void stack_destroy(stack_t *s) {
    free(s->values);
    pthread_mutex_destroy(&s->m);
    free(s);
}
```

```c
void push(stack_t *s, double v){
   pthread_mutex_lock(&s->m);
   s->values[(s->count)++] = v;
   pthread_mutex_unlock(&s->m);
}
double pop(stack_t *s){
   pthread_mutex_lock(&s->m);
   double v = s->values[--(s->count)];
   pthread_mutex_unlock(&s->m);
   return v;
}
int is_empty(stack_t *s){
   pthread_mutex_lock(&s->m);
   int result = s->count == 0;
   pthread_mutex_unlock(&s->m);
   return result;
}
```

```c
int main(){
   stack_t *s1 = stack_create(10 /* Max capacity*/);
   stack_t *s2 = stack_create(10);
   push(s1, 3.141);
   push(s2, pop(s1));
   stack_destroy(s2);
   stack_destroy(s1);
}
```

We still need to control overflow/underflow in pop and push

This can be done by using semaphores

With semaphores

```
#define SPACES 10
pthread_mutex_t m=
PTHREAD_MUTEX_INITIALIZER;
int count = 0;
double values[SPACES];
sem_t sitems, sremain;

void init() {
   sem_init(&sitems, 0, 0);
   sem_init(&sremains, 0, SPACES);
}
```

```
double pop() {
   /* Wait until there's
       at least one item*/
   sem_wait(&sitems);
   pthread_mutex_lock(&m);
   /* CRITICAL SECTION */
   double v = values[--count];
   pthread_mutex_unlock(&m);

   /*there's at least one space*/
   sem_post(&sremain);
   return v;
}
```

```
void push(double v) {
    /* Wait until there's at least one space*/
    sem_wait(&sremain);
    pthread_mutex_lock(&m);
    /* CRITICAL SECTION */
    values[count++] = v;
    pthread_mutex_unlock(&m);
    sem_post(&sitems); /* Hey world, there's at least one item*/
}
```

Exercise

How to modify the previous stack example for processes?

- Instead of mutex_lock, you need to use binary semaphore..

Implementing with condition variable: WRONG!

```
void push(stack_t *s, double v) {
 pthread_mutex_lock(&s->m);
 if(s->count == 0) pthread_cond_wait(&s->cv, &s->m);
 s->values[(s->count)++] = v;
 pthread_mutex_unlock(&s->m);
}

double pop(stack_t *s) {
 pthread_mutex_lock(&s->m);
 if(s->count == 0) pthread_cond_wait(&s->cv, &s->m);
 double v = s->values[--(s->count)];
 pthread_mutex_unlock(&s->m);
 return v;
}
```

Errors?

1. The first one is a simple one. In push, our check should be against the total capacity, not zero.
2. We only have if statement checks. wait() could spuriously wake up
3. We never signal any of the threads! Threads could get stuck waiting indefinitely.

Implementing with condition variable

```
void push(stack_t *s, double v) {
 pthread_mutex_lock(&s->m);
 while(s->count == capacity) pthread_cond_wait(&s->cv, &s->m);
 s->values[(s->count)++] = v;
 pthread_cond_signal(&s->cv);
 pthread_mutex_unlock(&s->m);

}
double pop(stack_t *s) {
 pthread_mutex_lock(&s->m);
 while(s->count == 0) pthread_cond_wait(&s->cv, &s->m);
 double v = s->values[--(s->count)];
 pthread_cond_broadcast(&s->cv);
 pthread_mutex_unlock(&s->m);
 return v;
}
```

- Signal wakes any thread
- Which might be another push thread

better to implement with two condition variable

See producer-consumer example

Implementing with 2 condition variables

```
void push(stack_t *s, double v) {
 pthread_mutex_lock(&s->m);
 while(s->count == capacity) pthread_cond_wait(&s->cv1, &s->m);
 s->values[(s->count)++] = v;
 pthread_cond_broadcast(&s->cv2);
 pthread_mutex_unlock(&s->m);

}
double pop(stack_t *s) {
 pthread_mutex_lock(&s->m);
 while(s->count == 0) pthread_cond_wait(&s->cv2, &s->m);
 double v = s->values[--(s->count)];
 pthread_cond_broadcast(&s->cv1);
 pthread_mutex_unlock(&s->m);
 return v;
}
```

Synchronization & deadlock

examples

- Reader-writer problem
- Review of producer-consumer problem
- Ring buffer (queue) implementation with condition variables & semaphores
- Deadlock, livelock
- Dining philosopher problem & different solutions

Reader-writer problem

Readers: only reads the data and does not update/change the data

Writers: both reads and writes the data

- allow multiple readers to access the data at the same time when there is no writing.
- To avoid data corruption, only one thread at a time may modify (write) the data structure and no readers may be reading at that time.

Problem: how can we efficiently synchronize multiple readers and writers such that multiple readers can read together, but a writer gets exclusive access?

First attempt with mutex

```
void read() {
   lock(&m)
   // do read stuff
   unlock(&m)
}

void write() {
   lock(&m)
   // do write stuff
   unlock(&m)
}
```

- No data corruption!
- But, only one reader can access the data. Readers wait for other readers

Second attempt

```
void read() {
   while(writing) {/*spin*/}

   reading = 1
   // do read stuff
   reading = 0
}
void write() {
   while(reading || writing) {/*spin*/}

   writing = 1
   // do write stuff
   writing = 0
}
```

- **Race condition: multiple readers and the writers at the same time**

With condition variables

Condition ok to read

Readers can access/read when no writers

Condition ok to write

Writers can access/update/write when no readers or writers

- ★ Only one thread should manipulate state variables (reading, writing) at a time.
  - ○ Critical section

Reader

```
read() {
    lock(&m)
    while (writing)
        cond_wait(&cv, &m)
    reading++;
    /* Read here! */
    reading--
    cond_signal(&cv)
    unlock(&m)
}
```

pthread_cond_wait performs 3 actions.

- it atomically unlocks the mutex
- and then sleeps
  - (until it is woken by pthread_cond_signal or pthread_cond_broadcast).
- **the awoken thread must re-acquire the mutex lock before returning.**

➔ Thus only one thread can actually be running inside the critical section.

Solution to Reader-Writer problem?

➔ if any writers are writing, a reader will enter the cond_wait.
- OK!

➔ Only one reader can read!
- because readers did not unlock the mutex.

A better version for multiple readers

```
read() {
    lock(&m)
    while (writing)
        cond_wait(&cv, &m)
    reading++;
    unlock(&m)

    /* Read here! */

    lock(&m)
    reading--
    cond_signal(&cv)
    unlock(&m)
}
```

Solution to Reader-Writer problem?

➔ if any writers are writing, a reader will enter the cond_wait.
- OK!

➔ only one thread can be executing code inside the critical section at any time!

➔ Readers can read at the same time

Does this mean that a writer and reader could read and write at the same time?

- No!
  - **cond_wait** requires the thread re-acquire the mutex lock before returning.

Writer

```
write() {
    lock(&m);
    while (reading || writing)
        cond_wait(&cv, &m);
    writing++;

    /* Write here! */

    writing--;
    cond_signal(&cv);//or cond_broadcast
    unlock(&m);
}
```

Explanation

- This uses pthread_cond_signal. This will only wake up one thread.
  - If many readers are waiting for the writer to complete, only one sleeping reader will be awoken from their slumber.
- The reader and writer should use cond_broadcast so that all threads should wake up and check their while-loop condition.

Problem with this solution

- Starvation: If readers are constantly arriving then a writer will never be able to proceed **(the 'reading' count never reduces to zero)**.

Last attempt

implement a bounded-wait for the writer.

- If a writer arrives they will still need to wait for existing readers
- however future readers must be placed in a “holding pen” and wait for the writer to finish.

The plan is when a writer arrives, and before waiting for current readers to finish, register our intent to write by incrementing a counter ‘writer’

Final writer solution

```
writer() {
    lock(&m)
    writers++
    while (reading || writing)
        cond_wait(&cv, &m)
    writing++
    unlock(&m)
    // perform writing here
    lock(&m)
    writing--
    writers--
    cond_broadcast(&cv)
    unlock(&m)
}
```

- Readers wait for the writers

final reader solution

```
read() {
   lock(&m)
   /*check if any writers waiting*/
   while(writers)
       cond_wait(&cv,&m)
   /*  No need to wait while(writing here) because it only exits the above loop when writing is zero*/
   while(writing)
       cond_wait(&cv,&m)
   reading++
   unlock(&m)
      /* Read here! */
   lock(&m)
   reading--
   cond_signal(&cv)
   unlock(&m)
}
```

- Readers wait for the writers
  - Writers have higher priority

Review: consumer-producer problem

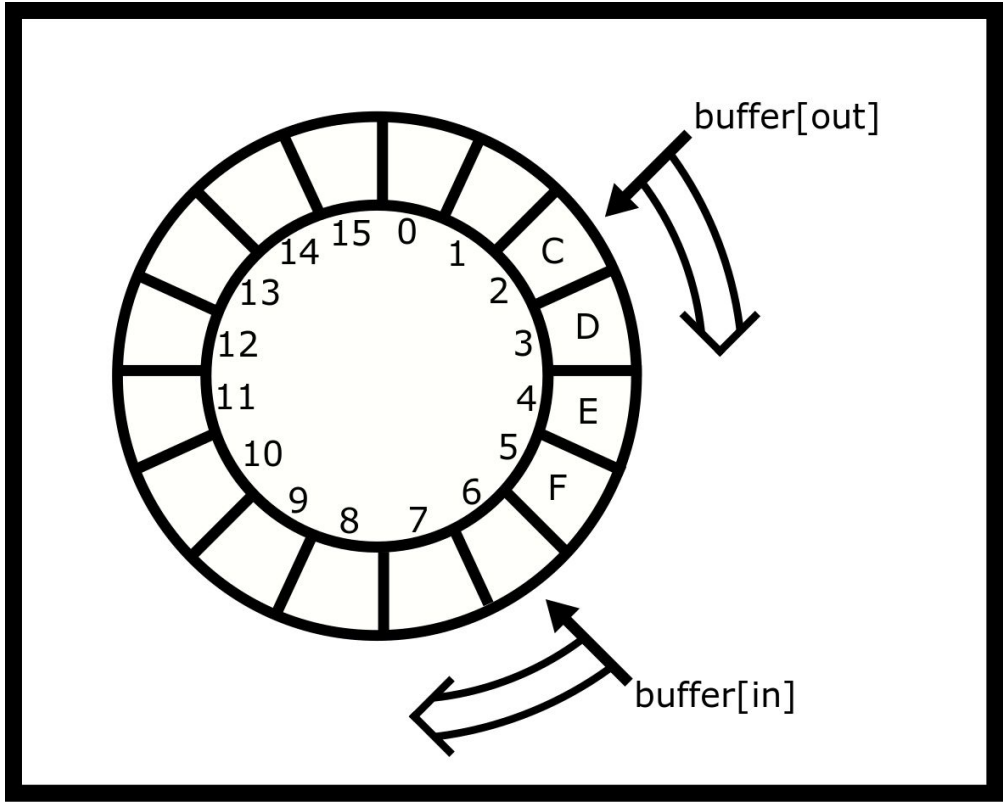
- a finite-size buffer
- and two classes of threads, producers and consumers,
  - producers: put items into the buffer
  - consumers: take items out of the buffer
- A producer must wait until the buffer has space before it can put something in,
- and a consumer must wait until something is in the buffer before it can take something out.

A **condition variable** represents a queue of threads waiting for some condition to be signaled.

Ring buffer

- A ring buffer is a simple, usually fixed-sized, storage mechanism
  - where contiguous memory is treated as if it is circular,
  - and two index counters keep track of the current beginning and end of the queue.
- As an array

```
void *buffer[16];
unsigned int in = 0, out = 0;
```



Ring buffer queue operations

```
void *buffer[16];
unsigned int in = 0, out = 0;

void enqueue(void *value) {
   buffer[in] = value;
   in++; /* Advance the index for next time */
   if (in == 16) in = 0; /* Wrap around! */
}
```

```
void *dequeue() {
   void *result = buffer[out];
   out++;
   if (out == 16) out = 0;
   return result;
}
```

- No control for buffer overflow/underflow!
- `if (in == 16) in = 0;`
  - `Using (in % 16) may cause problems if in is too big`
  - `b[ (in++) & (N-1) ] = value;`

```
#define N 16
void *b[N]
int in = 0, out = 0;
p_m_t lock; sem_t s1,s2;
void init() {
   p_m_init(&lock, NULL);
   sem_init(&s1, 0, 16);
   sem_init(&s2, 0, 0);
}
enqueue(void *value) {
   p_m_lock(&lock)
   sem_wait( &s1 )
   b[ (in++) & (N-1) ] = value
   sem_post(&s1)
   p_m_unlock(&lock)
}
```

Multithread solution: **incorrect** attempt

- The enqueue method waits and posts on the same semaphore (s1) and similarly with enqueue and (s2)
  - we decrement the value and then immediately increment the value,
  - so by the end of the function the **semaphore value is unchanged!**

- The initial value of s1 is 16,
  - the semaphore will never be reduced to zero
    - enqueue will not block if the ring buffer is full
    - so overflow is possible.

```
void init() {
   p_m_init(&lock, NULL);
   sem_init(&s1, 0, 16);
   sem_init(&s2, 0, 0);
}
void *dequeue(){
   p_m_lock(&lock)
   sem_wait(&s2)
   void *result = b[(out++) & (N-1) ]
   sem_post(&s2)
   p_m_unlock(&lock)
   return result
}
```

- The initial value of s2 is zero, so calls to dequeue will always block and never return!

- The order of mutex lock and sem_wait will need to be swapped; however, this example is so broken that this bug has no effect!

Another **incorrect** attempt

```
void init() {
   sem_init(&s1,0,16);  sem_init(&s2,0,0);
}
enqueue(void *value){
   sem_wait(&s2);
   p_m_lock(&lock);
   b[ (in++) & (N-1) ] = value;
   p_m_unlock(&lock);
   sem_post(&s1);
}
void *dequeue(){
   sem_wait(&s1);
   p_m_lock(&lock)
   void *result = b[(out++) & (N-1)];
   p_m_unlock(&lock);
   sem_post(&s2);
   return result;
}
```

- The initial value of s2 is 0. **Enqueue will block on the first call** to sem_wait even though the buffer is empty!
- The initial value of s1 is 16. Dequeue **will not block on the first call** to sem_wait even though the buffer is empty - Underflow! The dequeue method will return invalid data.
- The code does not satisfy Mutual Exclusion. Two threads can modify in or out at the same time! The code appears to use mutex lock. Unfortunately, the lock was never initialized with pthread_mutex_init() or PTHREAD_MUTEX_INITIALIZER - so the lock may not work (pthread_mutex_lock may simply do nothing)

Correct implementation: enqueue

```
#include <pthread.h>
#include <semaphore.h>
#define N (16) // N must be 2^i
void *b[N]
int in = 0, out = 0
p_m_t lock = PTHREAD_MUTEX_INITIALIZER;
sem_t countsem, spacesem;
void init() {
   sem_init(&countsem, 0, 0);
   sem_init(&spacesem, 0, 16);
}
```

```
enqueue(void *value){
   /* wait if there is no space left:*/
   sem_wait( &spacesem )

   p_m_lock(&lock)
   b[ (in++) & (N-1) ] = value
   p_m_unlock(&lock)

   /* increment the count of the number of items*/
   sem_post(&countsem)
}
```

Correct implementation: dequeue

```
#include <pthread.h>
#include <semaphore.h>
#define N (16) // N must be 2^i
void *b[N]
int in = 0, out = 0
p_m_t lock = PTHREAD_MUTEX_INITIALIZER;
sem_t countsem, spacesem;
void init() {
   sem_init(&countsem, 0, 0);
   sem_init(&spacesem, 0, 16);
}
```

```
void *dequeue(){
   /* Wait if there are no items in the buffer*/
   sem_wait(&countsem);

   p_m_lock(&lock);
   void *result = b[(out++) & (N-1)];
   p_m_unlock(&lock);

   /* Increment the count of the number of spaces*/
   sem_post(&spacesem);
   return result
}
```

Exercises

- What would happen if the order of **sem_wait** and **pthread_mutex_lock** calls were swapped?

- What would happen if the order of **pthread_mutex_unlock** and **sem_post** calls were swapped?

```
void *dequeue(){
   /* Wait if there are no items in the buffer*/
   sem_wait(&countsem);
   p_m_lock(&lock);
   void *result = b[(out++) & (N-1)];
   p_m_unlock(&lock);
   /* Increment the count of the number of
spaces*/
   sem_post(&spacesem);
   return result
}
```

Ring buffer as a consumer-producer with cvs

```
typedef struct {
   char buf[BSIZE];
   int occupied;
   int in;              /*index of in*/
   int out;             /*index of out*/
   pthread_mutex_t mutex;
   pthread_cond_t more; /*there is item*/
   pthread_cond_t less;    /*there is room*/
} buffer_t;
```

```
void enqueue(buffer_t *b, char item){
   pthread_mutex_lock(&b->mutex);
   while (b->occupied >= BSIZE)
       pthread_cond_wait(&b->less, &b->mutex);
   assert(b->occupied < BSIZE);
   b->buf[b->in++] = item;
   b->in &= BSIZE-1;
   b->occupied++;
   pthread_cond_signal(&b->more);
   pthread_mutex_unlock(&b->mutex);
}
```

https://docs.oracle.com/cd/E19455-01/806-5257/6je9h032r/index.html#sync-ex-33

```
char dequeue(buffer_t *b){
   char item;
   pthread_mutex_lock(&b->mutex);
   while(b->occupied <= 0){
       pthread_cond_wait(&b->more, &b->mutex);
   }
   item = b->buf[b->out++];
   b->out &= BSIZE-1;
   b->occupied--;
   pthread_cond_signal(&b->less);
   pthread_mutex_unlock(&b->mutex);
   return(item);
}
```

Dequeue is a consumer

Enqueue is a producer

https://docs.oracle.com/cd/E19455-01/806-5257/6je9h032r/index.html#sync-ex-33

Deadlock problem

Deadlock

Deadlock describes a situation where two or more threads are blocked forever, waiting for each other.

Deadlock example

```
mutex_t m1, m2;
void p1(void *ignored) {
    lock(m1);
    lock(m2);
    /* critical section */
    unlock(m2);
    unlock(m1);
}
```

```
void p2(void *ignored) {
    lock(m2);
    lock(m1);
    /* critical section */
    unlock(m1);
    unlock(m2);
}
```

This program can cease to make progress – how?
Can you have deadlock w/o mutexes?

https://www.scs.stanford.edu/23wi-cs212/notes/synchronization2.pdf

Livelock

A thread often acts in response to the action of another thread. If the other thread's action is also a response to the action of another thread, then *livelock* may result.

Starvation

Starvation describes a situation where a thread is unable to gain regular access to shared resources and is unable to make progress.

Deadlock conditions (Coffman conditions)

1. Limited access (mutual exclusion):

- Resource can only be shared with finite users

2. No preemption:

- Once resource granted, cannot be taken away

3. Multiple independent requests (hold and wait):

- Don’t ask all at once (wait for next resource while holding current one)

4. Circularity in graph of requests

See also simulations in Deadlock - Wikipedia

All of 1–4 necessary for deadlock to occur

Two approaches to dealing with deadlock:

- Pro-active: prevention

- Reactive: detection + corrective action

https://www.scs.stanford.edu/23wi-cs212/notes/synchronization2.pdf

Prevent by eliminating one condition

1. Limited access (mutual exclusion):

- Resource can only be shared with finite users

2. No preemption:

- Once resource granted, cannot be taken away

3. Multiple independent requests (hold and wait):

- Don’t ask all at once (wait for next resource while holding current one)

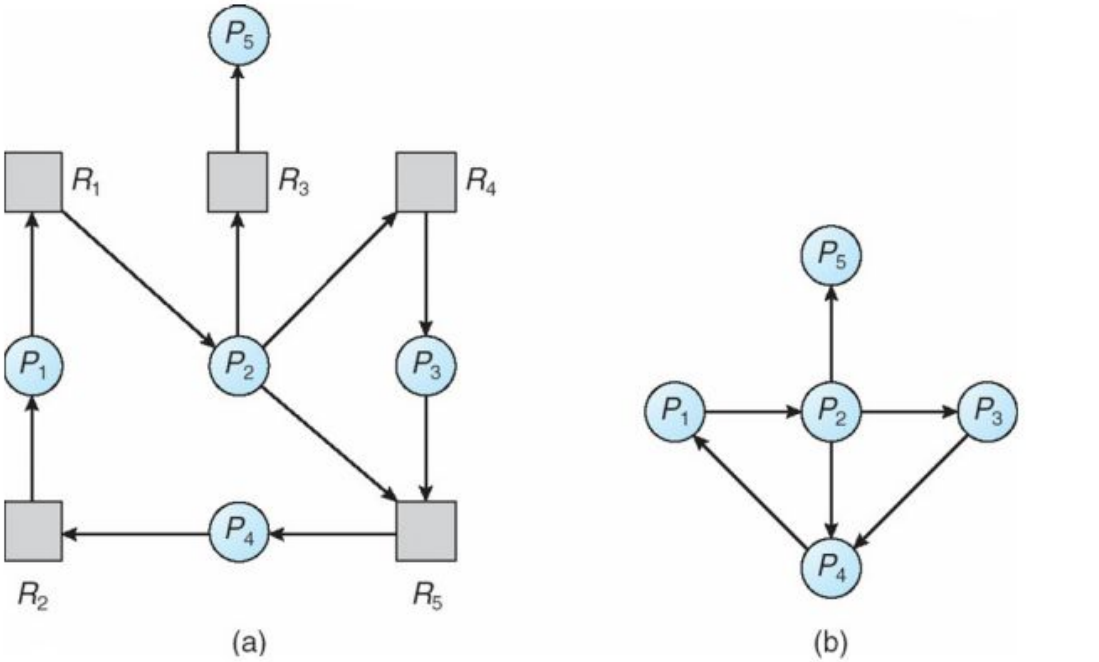
4. Circularity in graph of requests

- **Single lock for entire system: (problems?)**
- **Partial ordering of resources (next)**

https://www.scs.stanford.edu/23wi-cs212/notes/synchronization2.pdf

Detecting deadlocks

- Static approaches (hard)
- Dynamically, program grinds to a halt
  - Threads package can diagnose by keeping track of locks held:



Resource-Allocation Graph Corresponding wait-for graph

https://www.scs.stanford.edu/23wi-cs212/notes/synchronization2.pdf

Fixing and debugging deadlocks

Ostrich algorithm: ignore the deadlock, prevent processes from context-switching

Reboot system / restart application

Examine hung process with debugger

Threads package can deduce partial order

- For each lock acquired, order with other locks held
- If cycle occurs, abort with error
- Detects potential deadlocks even if they do not occur

Or use transactions. . .

- - Another paradigm for handling concurrency
- - Often provided by databases, but some OSes use them

https://www.scs.stanford.edu/23wi-cs212/notes/synchronization2.pdf

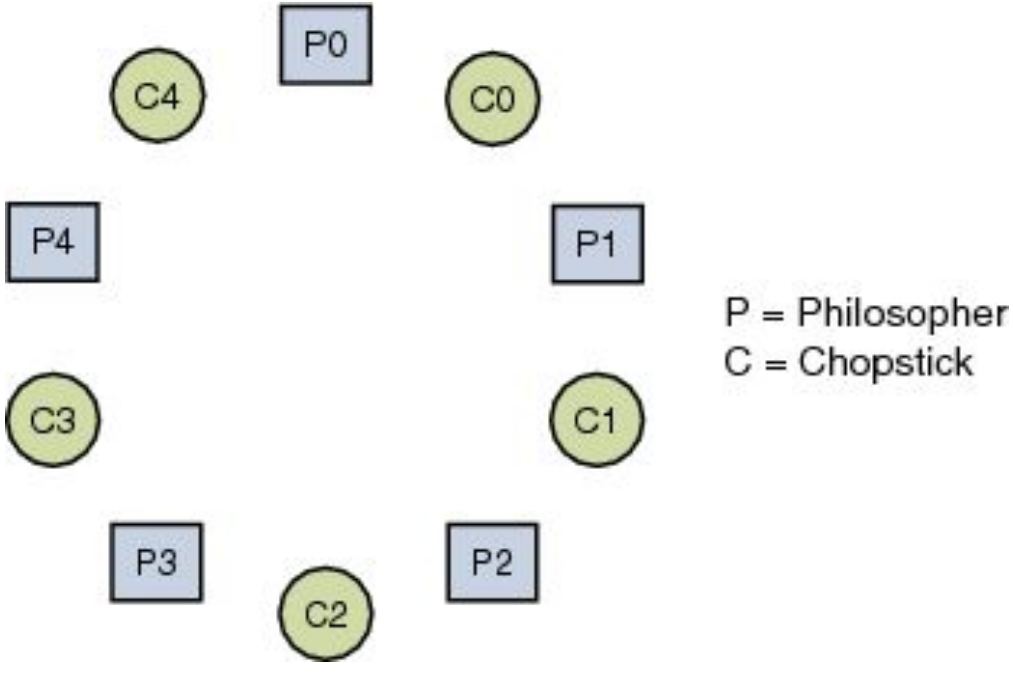
Algorithmic Approaches to solve deadlock

Banker’s algorithm

Banker's algorithm - Wikipedia

- How much of each resource each process could possibly request ("MAX")
- How much of each resource each process is currently holding ("ALLOCATED")
- How much of each resource the system currently has available ("AVAILABLE")

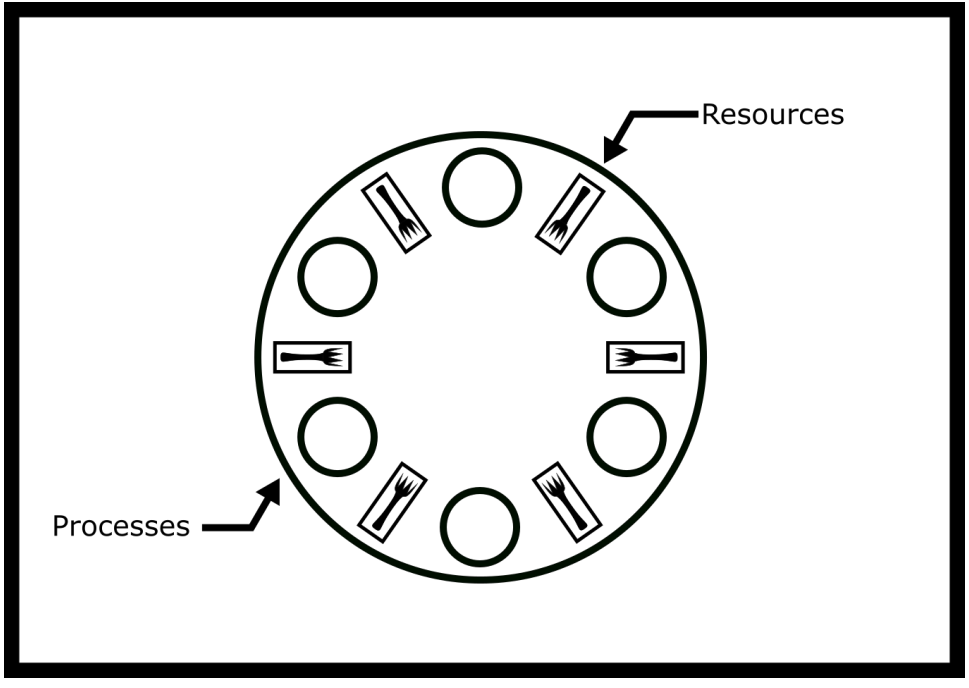
Deadlock example with dining philosopher problem



Chapter 3 The Deadlock Tutorial (Sun Studio 12: Thread Analyzer User's Guide)

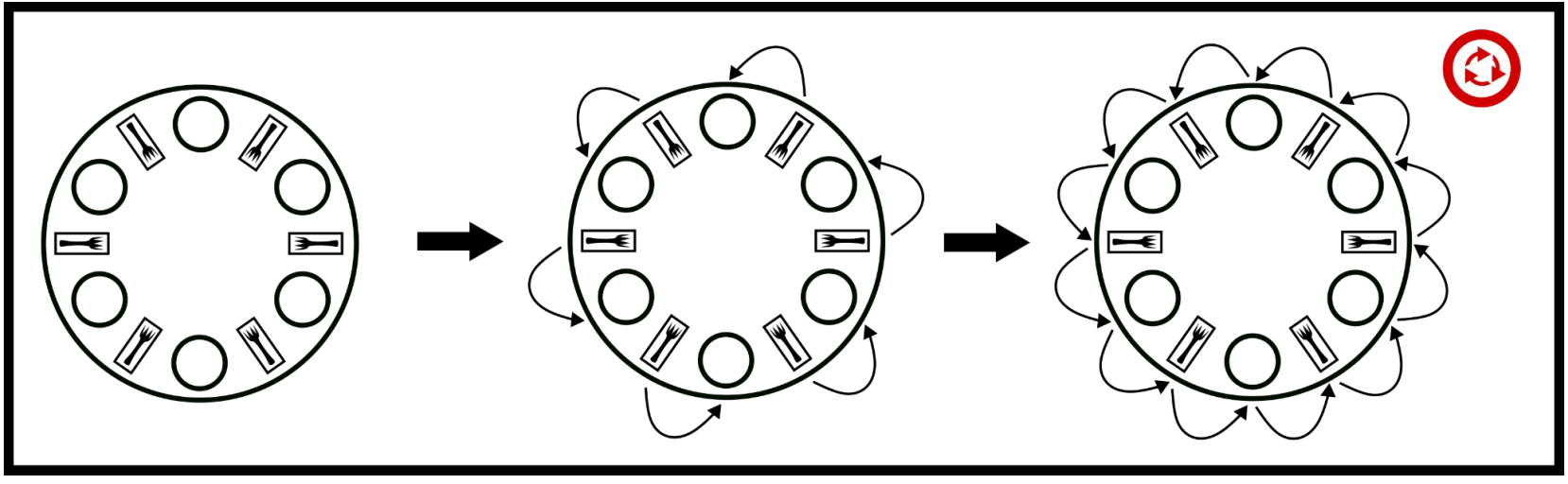
Dining philosopher problem

```
philosopher(){
     while(true){
          think()
          pick(left_chopstick)
          pick(right_chopstick)
          eat()
          putdown_chopsticks()
     }
}
```



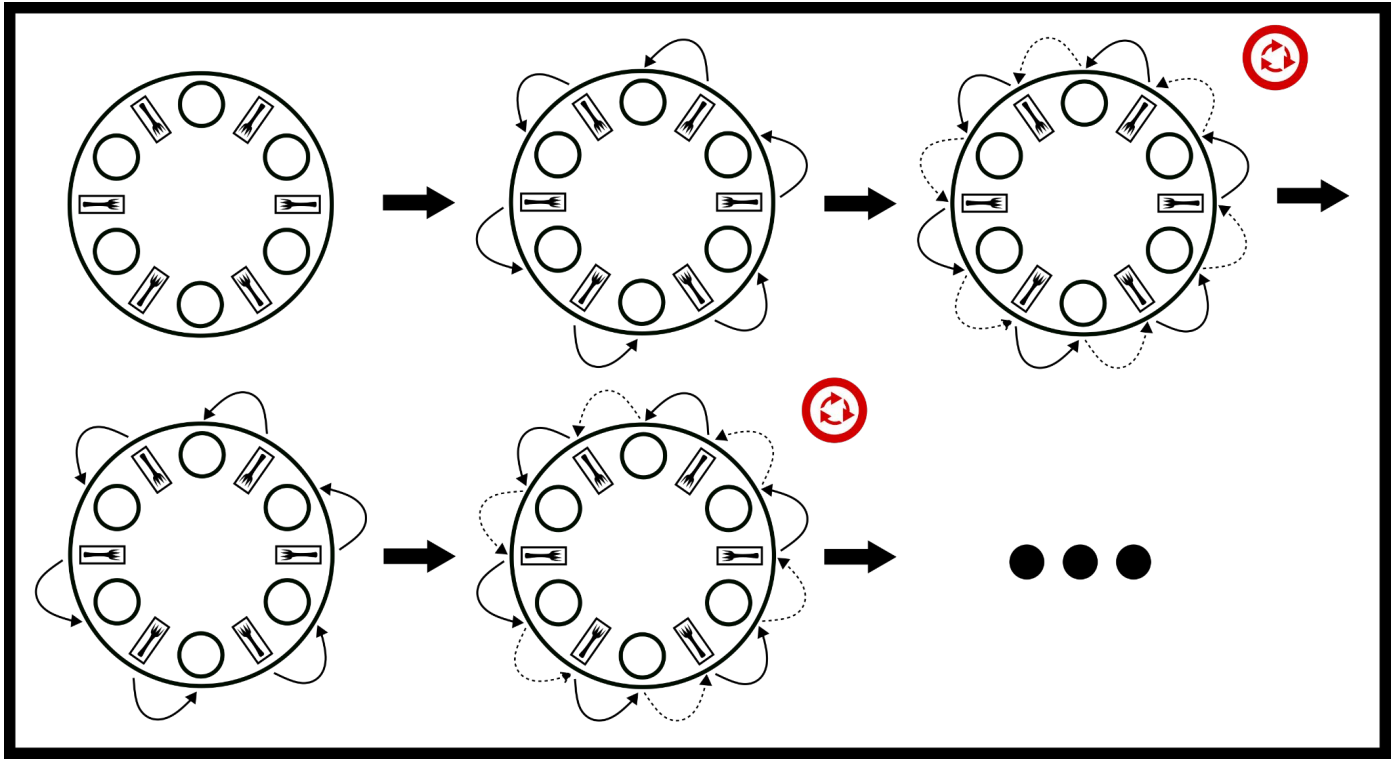
How deadlock occurs?

What if everyone picks up their left chopstick and is waiting on their right chopstick?



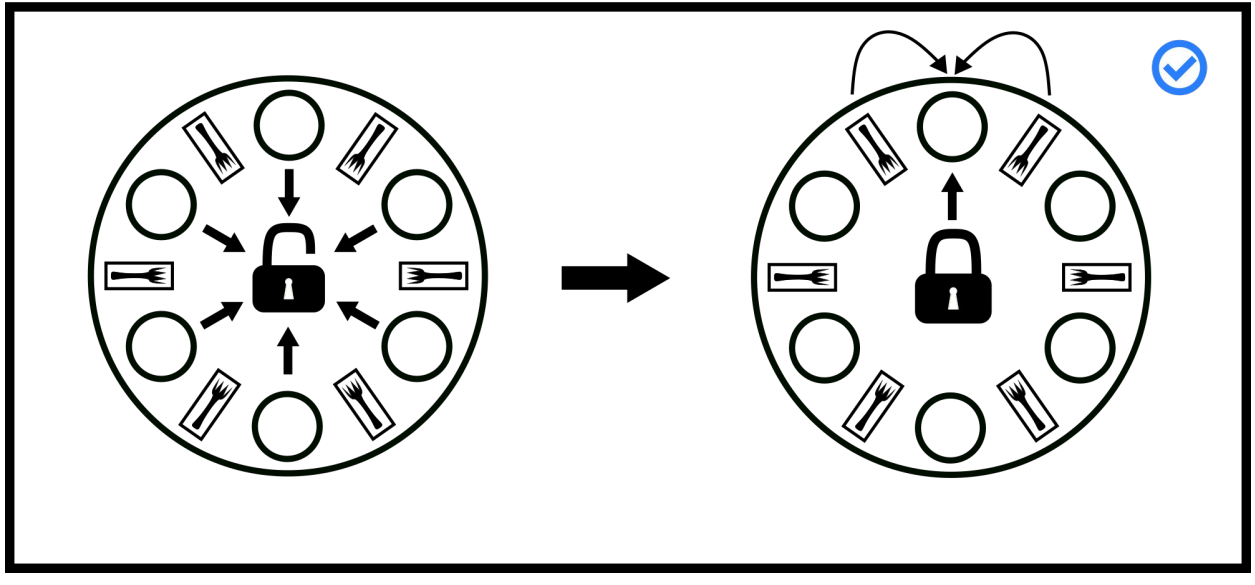
resource allocation graph that shows how the system could be deadlocked

LIVELOCK: a time evolution of the break hold and wait would look like.



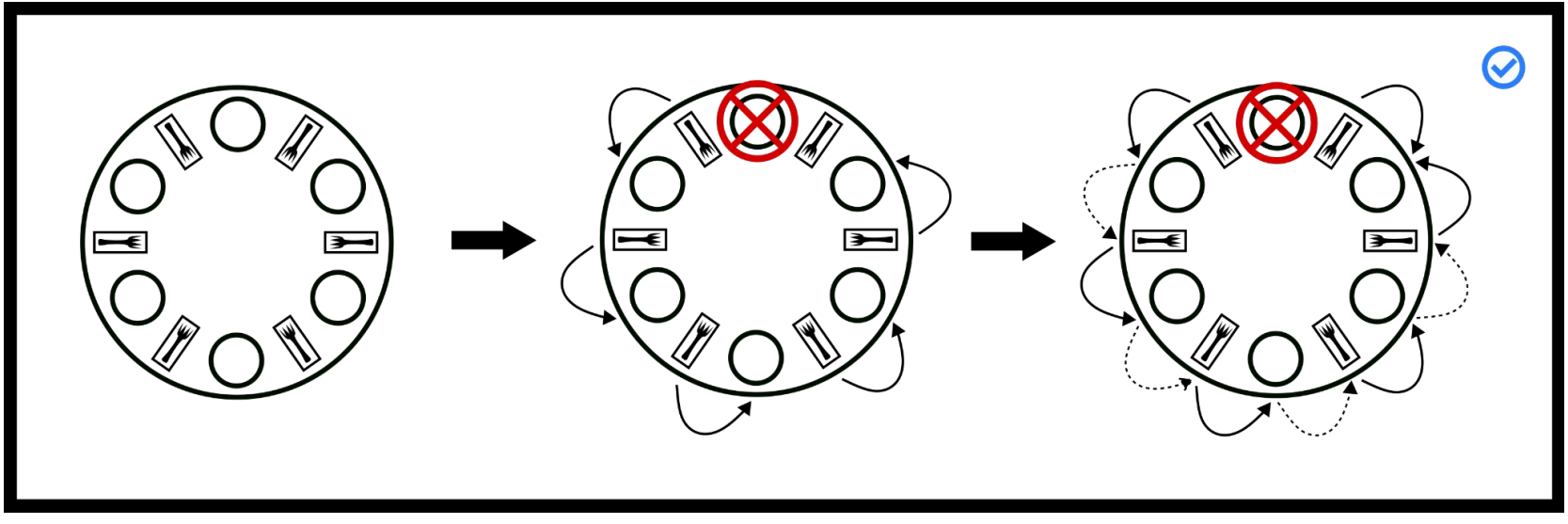
Arbitrator solution

- A philosopher can only pick up both or none by introducing an arbitrator, e.g., a waiter
  - Waiter can be implemented as a mutex
- Works but parallelism is reduced
  - All philosophers wait for waiter



Stalling solution (leaving table)

Remove philosophers until deadlock is not possible. No circular wait means deadlock is not possible.



Partial (hierarchical) ordering (Dijkstra's solution)

Each process picks only lower numbered chopstick first, then the higher numbered chopstick.